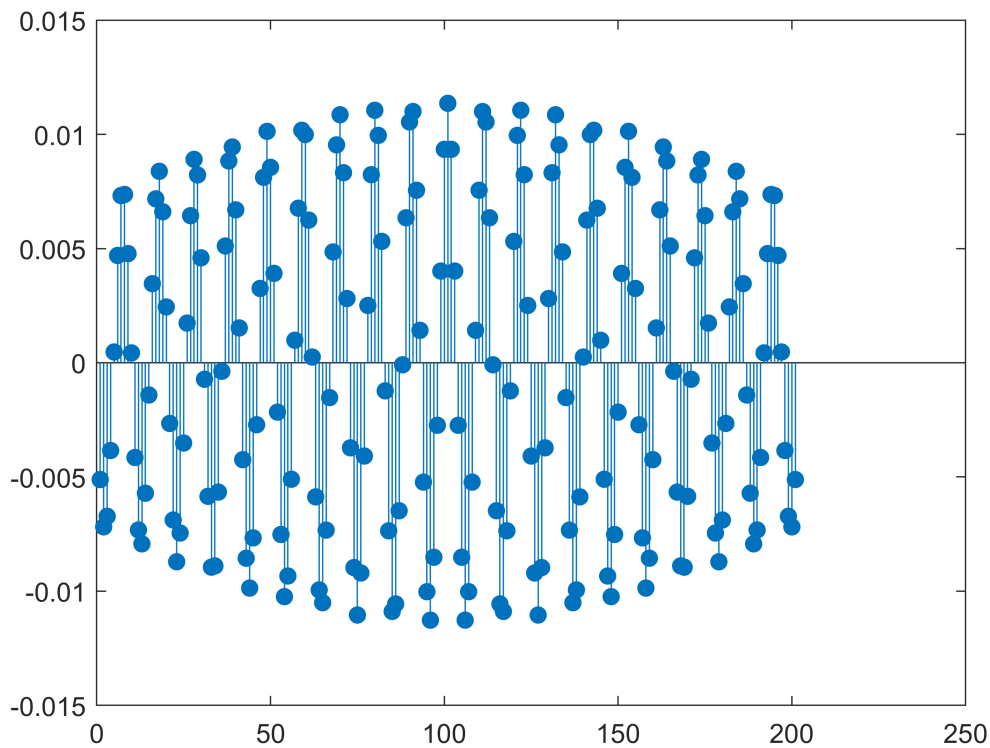
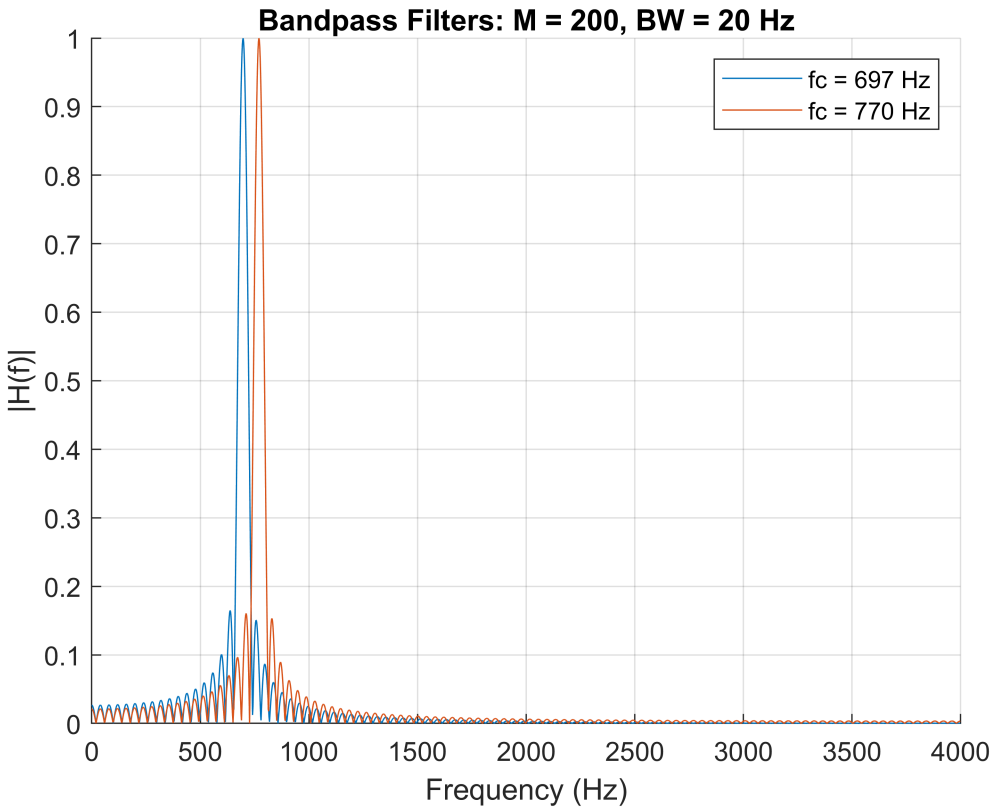
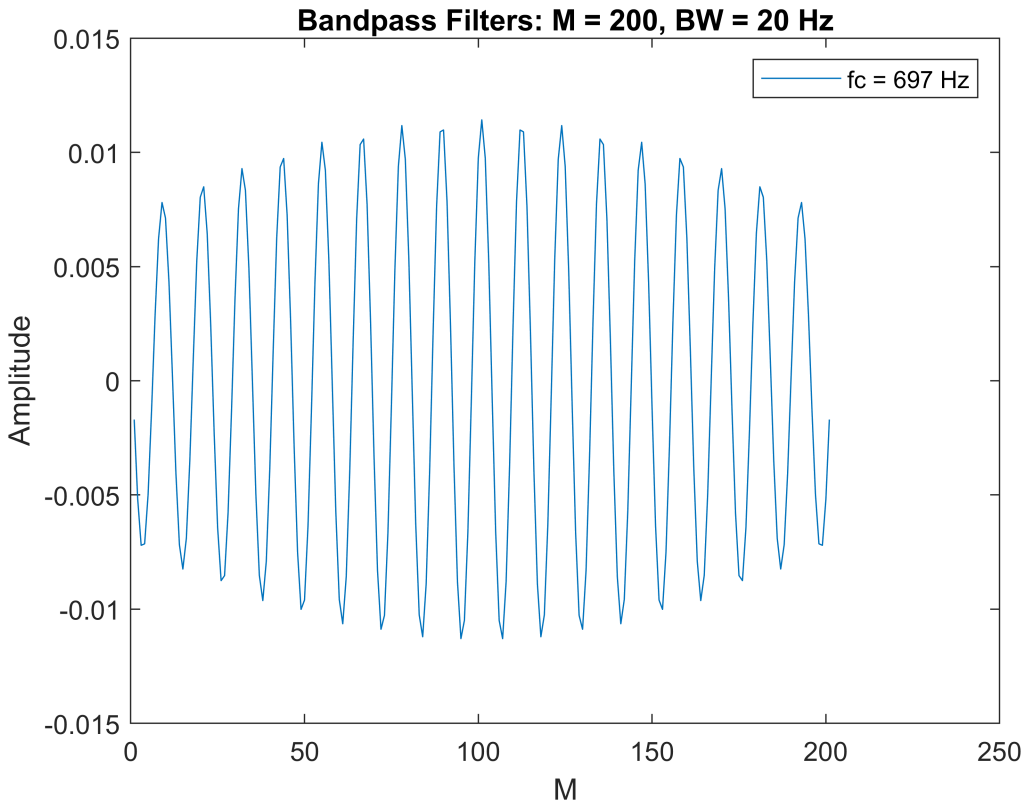
```
fs = 8000;
fc1 = 697;
fc2 = 770;
bw = 20;
M=200;

Wn0 = [fc1-bw, fc1+bw]/(fs/2);
Wn1 = [fc2-bw, fc2+bw]/(fs/2);

b1=fir1(M, Wn0, rectwin(M+1));
b2=fir1(M, Wn1, rectwin(M+1));
figure;
stem(b2, 'filled')
```



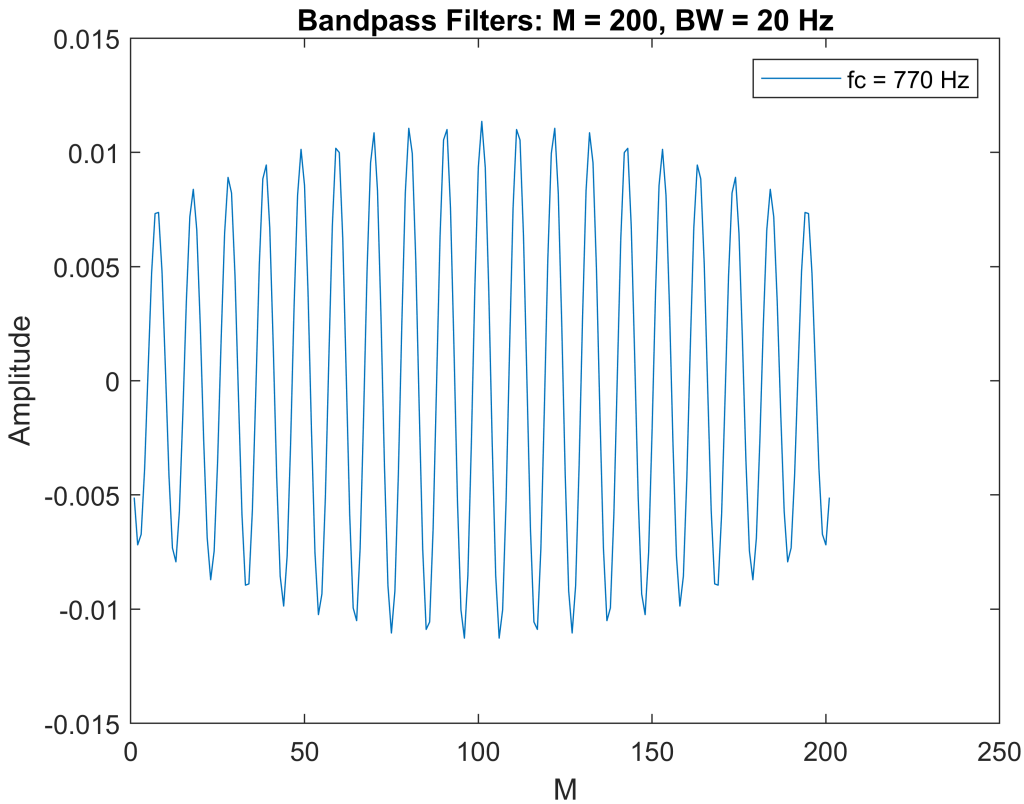

```
figure;

Np = 2048;
[H1, fv] = freqz(b1,1,Np,fs);
[H2, fv] = freqz(b2,1,Np,fs);
hold on
plot(fv, abs(H1))
plot(fv, abs(H2))
grid on
hold off
title(sprintf('Bandpass Filters: M = %d, BW = %d Hz', M, bw));
xlabel('Frequency (Hz)');
ylabel('|H(f)|');
legend(sprintf('fc = %d Hz', fc1), sprintf('fc = %d Hz', fc2));
```



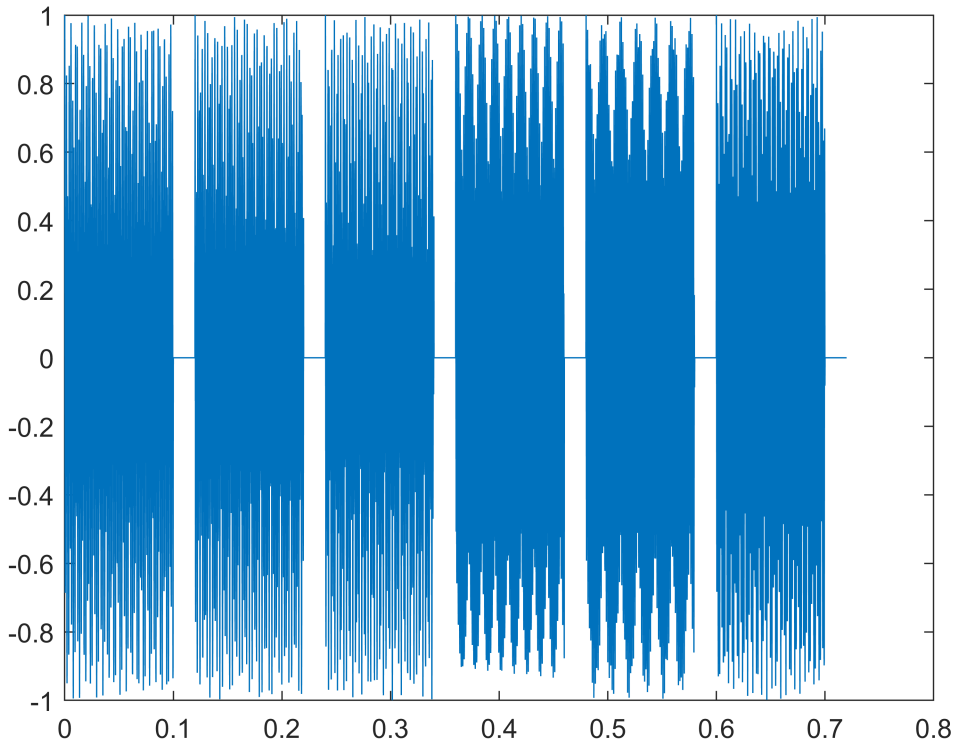

```
plot(b1)
title(sprintf('Bandpass Filters: M = %d, BW = %d Hz', M, bw));
xlabel('M');
ylabel('Amplitude');
legend(sprintf('fc = %d Hz', fc1));
```




```
plot(b2)
title(sprintf('Bandpass Filters: M = %d, BW = %d Hz', M, bw));
xlabel('M');
ylabel('Amplitude');
legend(sprintf('fc = %d Hz', fc2));
```

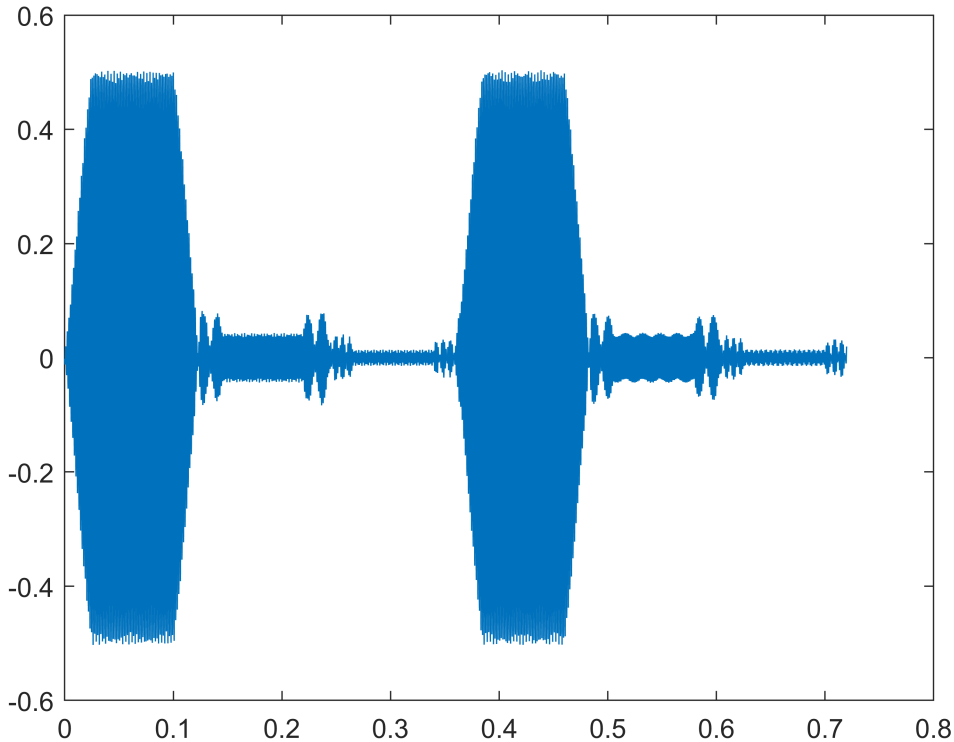



```
tone_len = .1;
quiet_len = .02;
Fs = 8000;
n = 6;
dT = 1/Fs;

x = my_dtmf(tone_len, quiet_len, Fs, [1, 4, 7, 3, 6, 9]);

time = n * (tone_len + quiet_len);

tvec = (0: (time * Fs -1) ) * dT;
plot(tvec, x)
```



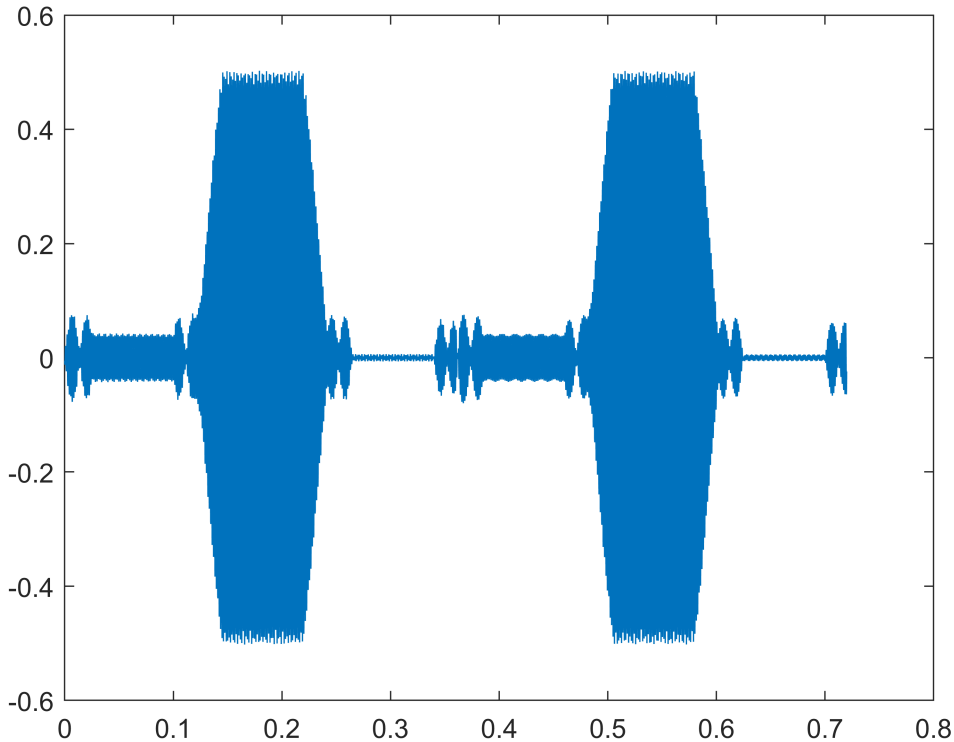

```
sound(x, Fs)
```

```
a=1;
y1 = filter(b1,a,x);
plot(tvec,y1);
```




```
sound(y1, Fs)
```

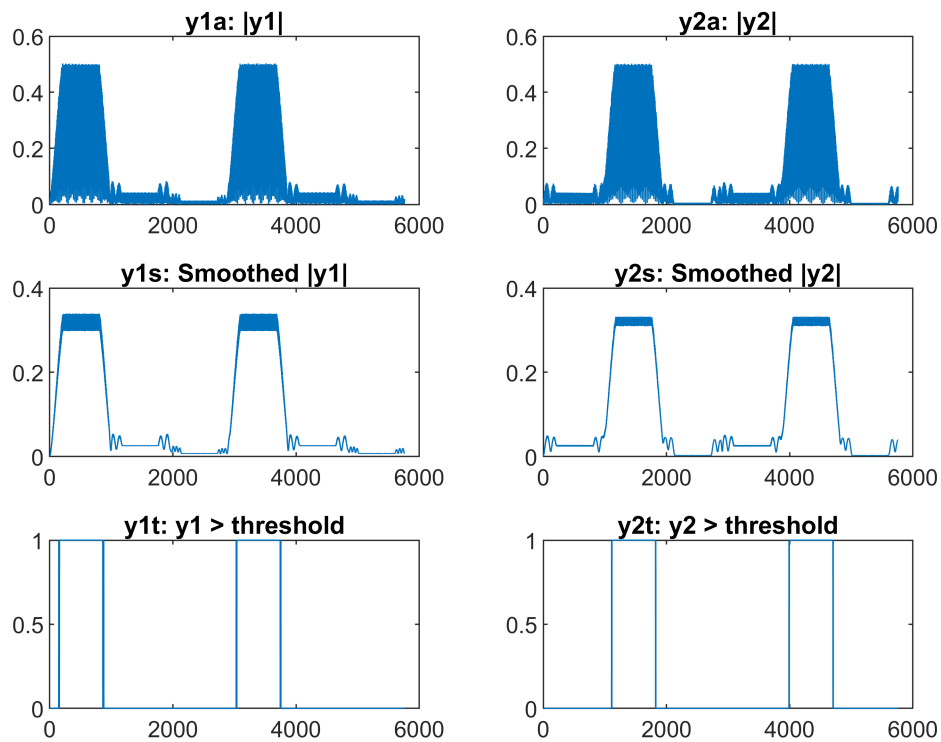
```
y2 = filter(b2,a,x);
plot(tvec,y2);
```

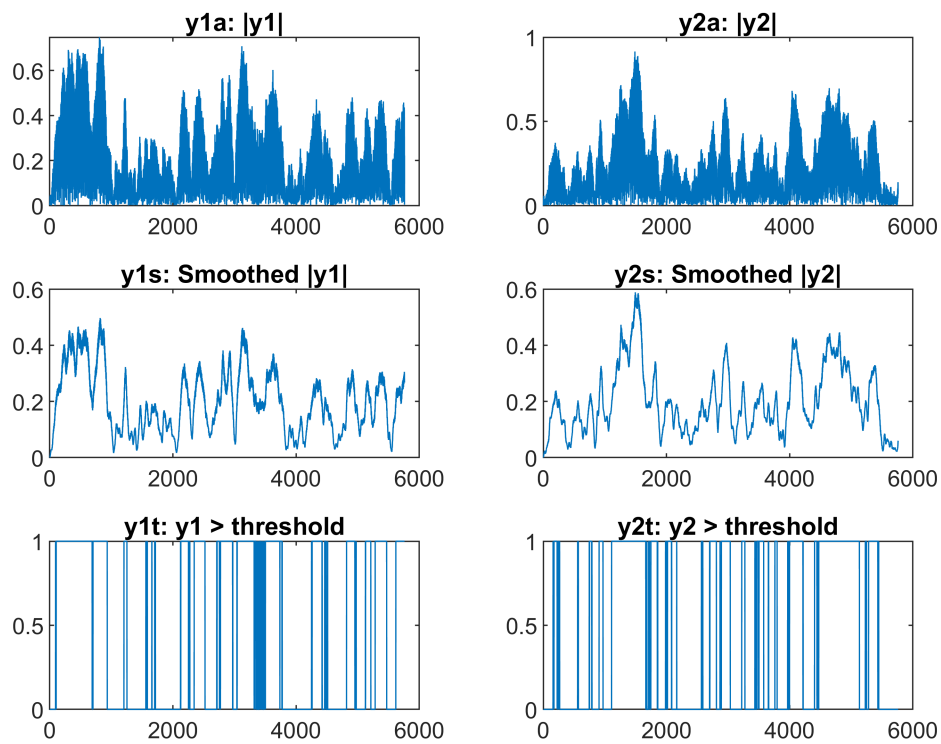



```
sound(y2, Fs)
```

```
y1a = abs(y1);
y2a = abs(y2);

smoothLength = 20;                  % Length of moving average filter
bma = ones(1, smoothLength) / smoothLength;  % Moving average filter


y1s = filter(bma, a, y1a);
y2s = filter(bma, a, y2a);

figure;

% Top row: Absolute values
subplot(3,2,1); plot(y1a); title('y1a: |y1|');
subplot(3,2,2); plot(y2a); title('y2a: |y2|');

% Middle row: Smoothed absolute values
subplot(3,2,3); plot(y1s); title('y1s: Smoothed |y1|');
subplot(3,2,4); plot(y2s); title('y2s: Smoothed |y2|');


lowThresh1 = .17;
highThresh1 = .3;
lowThresh2 = .17;
highThresh2 = .3;
```

```
thresh1 = (lowThresh1 + highThresh1) / 2;
thresh2 = (lowThresh2 + highThresh2) / 2;
y1t = y1s > thresh1;
y2t = y2s > thresh2;

% Bottom row: Thresholded logical signals
subplot(3,2,5); plot(y1t); title('y1t: y1 > threshold');
subplot(3,2,6); plot(y2t); title('y2t: y2 > threshold');
```




```
testsign = x + 2*randn(1, length(x));
sound(testsign)
a=1;

y1 = filter(b1,a,testsign);
y2 = filter(b2,a,testsign);

y1a = abs(y1);
y2a = abs(y2);

smoothLength = 20;                  % Length of moving average filter
bma = ones(1, smoothLength) / smoothLength;  % Moving average filter


y1s = filter(bma, a, y1a);
y2s = filter(bma, a, y2a);

figure;
```

```
% Top row: Absolute values
subplot(3,2,1); plot(y1a); title('y1a: |y1|');
subplot(3,2,2); plot(y2a); title('y2a: |y2|');

% Middle row: Smoothed absolute values
subplot(3,2,3); plot(y1s); title('y1s: Smoothed |y1|');
subplot(3,2,4); plot(y2s); title('y2s: Smoothed |y2|');


lowThresh1 = .23;
highThresh1 = .15;
lowThresh2 = .23;
highThresh2 = .15;
thresh1 = (lowThresh1 + highThresh1) / 2;
thresh2 = (lowThresh2 + highThresh2) / 2;
y1t = y1s > thresh1;
y2t = y2s > thresh2;

% Bottom row: Thresholded logical signals
subplot(3,2,5); plot(y1t); title('y1t: y1 > threshold');
subplot(3,2,6); plot(y2t); title('y2t: y2 > threshold');
```